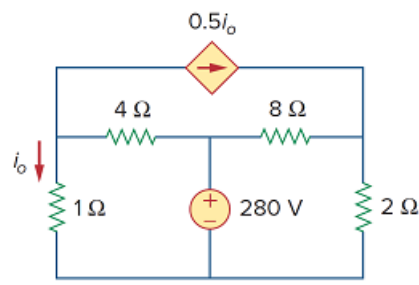


Problem

Calculate the power dissipated in each resistor in the circuit of Fig. 3.104.

Figure 3.104:



Step-by-step solution

Step 1 of 7

Refer to circuit diagram in Figure 3.104 in the text book.

Identify loops and currents in the circuit diagram shown in Figure 1.

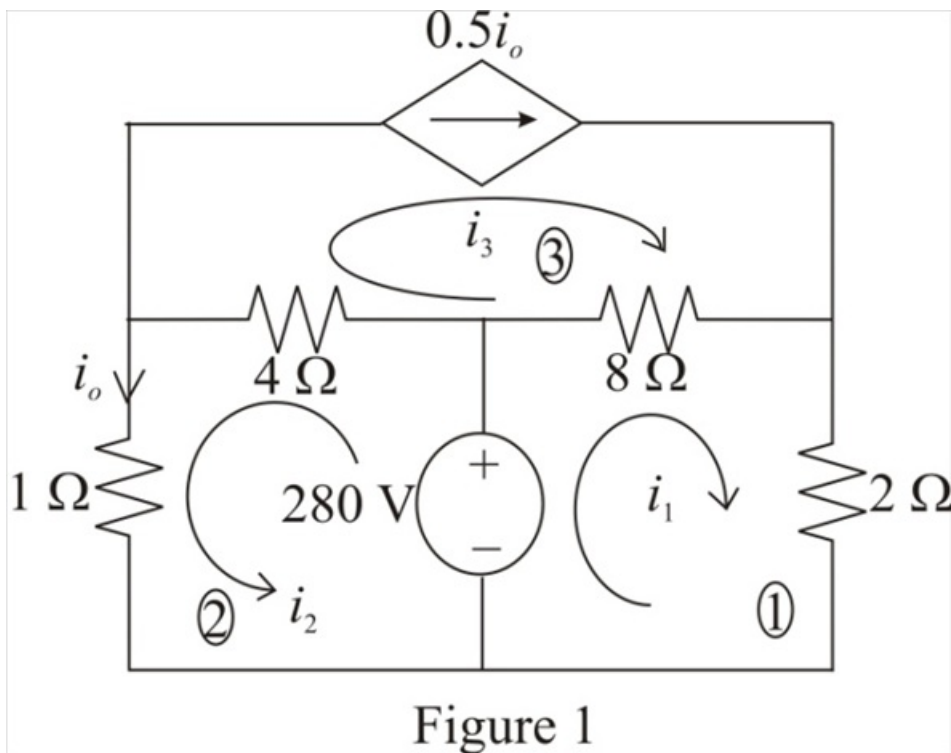


Figure 1

Step 2 of 7

From the current,

$$i_2 = i_o$$

And,

$$i_3 = 0.5i_o$$

Apply Kirchhoff's Voltage Law to Loop-1.

$$-280 + (i_1 - i_3)(8) + i_1(2) = 0$$

Substitute $0.5i_o$ is i_3 .

$$-280 + (i_1 - 0.5i_o)(8) + i_1(2) = 0$$

$$-280 + 10i_1 - 4i_o = 0$$

$$10i_1 = 4i_o + 280$$

$$i_1 = \frac{4i_o + 280}{10} \dots\dots (1)$$

Step 3 of 7

Apply Kirchhoff's Voltage Law to Loop-1.

$$-280 + (i_2 + i_3)(4) + i_2(1) = 0$$

Substitute $0.5i_o$ is i_3 and i_o is i_2 .

$$-280 + (i_o + 0.5i_o)(4) + i_o(1) = 0$$

$$-280 + (1.5i_o)(4) + i_o(1) = 0$$

$$7i_o = 280$$

$$i_o = 40 \text{ A}$$

Recall equation (1).

$$i_1 = \frac{4i_o + 280}{10}$$

Substitute 40 A for i_o .

$$i_1 = \frac{4(40) + 280}{10}$$

$$= \frac{440}{10}$$

$$= 44 \text{ A}$$

[Comments \(1\)](#)



Anonymous

this is loop 1

Step 4 of 7

Calculate the power absorbed by 2Ω resistor.

$$P_2 = (i_1)^2(2)$$

Substitute 44 A for i_1 .

$$P_2 = (44)^2(2)$$

$$= (1936)(2)$$

$$= 3872 \text{ W}$$

Therefore, the value of the power absorbed by 2Ω resistor is 3872 W .

Step 5 of 7

Calculate the power absorbed by $8\ \Omega$ resistor.

$$P_8 = (i_1 - i_3)^2 (8)$$

Substitute $0.5i_o$ is i_3 .

$$P_8 = (i_1 - 0.5i_o)^2 (8)$$

Substitute 44 A for i_1 and 40 A for i_o .

$$\begin{aligned} P_8 &= (44 - 0.5(40))^2 (8) \\ &= (44 - 10)^2 (8) \\ &= (24)^2 (8) \\ &= (576)(8) \end{aligned}$$

$$= 4608\text{ W}$$

Therefore, the value of the power absorbed by $8\ \Omega$ resistor is, 4608 W .

[Comments \(1\)](#)



Anonymous

$-0.5(40)=20$ not 10 . But the answer is right.

Step 6 of 7

Calculate the power absorbed by $4\ \Omega$ resistor.

$$P_4 = (i_2 + i_3)^2 (4)$$

Substitute $0.5i_o$ is i_3 and i_o is i_2 .

$$\begin{aligned} P_4 &= (i_o + 0.5i_o)^2 (4) \\ &= (1.5i_o)^2 (4) \end{aligned}$$

Substitute 40 A for i_o .

$$\begin{aligned} P_4 &= [(1.5)(40)]^2 (4) \\ &= (60)^2 (4) \\ &= (3600)(4) \\ &= 14400\text{ W} \end{aligned}$$

Therefore, the value of the power absorbed by $4\ \Omega$ resistor is, 14400 W .

Step 7 of 7

Calculate the power absorbed by $1\ \Omega$ resistor.

$$P_1 = (i_o)^2 (1)$$

Substitute 40 A for i_o .

$$\begin{aligned} P_1 &= (40)^2 (1) \\ &= 1600\text{ W} \end{aligned}$$

Therefore, the value of the power absorbed by $1\ \Omega$ resistor is, 1600 W .